



Summary Statistics

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 Stat 141 | Fall 2026
 Wednesday, Week 2

Announcements

- Data Lab at Reed!
 - Many resources including drop-in help, online R tutorials
- Math/Stats colloquium tomorrow! Come and hear about examples of stats research!

Reminders

- Test rendering your HW submission early
- Select pages for each problem part on Gradescope to receive full credit

Last Time

- Learned about the structure of **ggplot2**
- Learn five standard graphs for **visualizing** data: histograms, boxplots, barplots, scatterplots, and linegraphs

Goals for Today

- Consider measures for **numerically summarizing** quantitative data
 - Center
 - Spread/variability
- Consider measures for **numerically summarizing** categorical data
 - Contingency tables

Import the Data

```
1 biketown <- read.csv("data/biketown.csv")
2
3 # Inspect the data
4 glimpse(biketown)
```

Rows: 9,999
 Columns: 19

\$ RouteID	<int> 4074085, 3719219, 3789757, 3576798, 3459987, 3947695,...
\$ PaymentPlan	<chr> "Subscriber", "Casual", "Casual", "Subscriber", "Casu...
\$ StartHub	<chr> "SE Elliott at Division", "SW Yamhill at Director Par...
\$ StartLatitude	<dbl> 45.50513, 45.51898, 45.52990, 45.52389, 45.53028, 45...
\$ StartLongitude	<dbl> -122.6534, -122.6813, -122.6628, -122.6722, -122.6547...
\$ StartDate	<chr> "8/17/2017", "7/22/2017", "7/27/2017", "7/12/2017", "..."
\$ StartTime	<chr> "10:44:00", "14:49:00", "14:13:00", "13:23:00", "19:3...
\$ EndHub	<chr> "Blues Fest - SW Waterfront at Clay - Disabled", "SW ...
\$ EndLatitude	<dbl> 45.51287, 45.52142, 45.55902, 45.53409, 45.52990, 45...
\$ EndLongitude	<dbl> -122.6749, -122.6726, -122.6355, -122.6949, -122.6628...
\$ EndDate	<chr> "8/17/2017", "7/22/2017", "7/27/2017", "7/12/2017", "..."
\$ EndTime	<chr> "10:56:00", "15:00:00", "14:42:00", "13:38:00", "20:3...

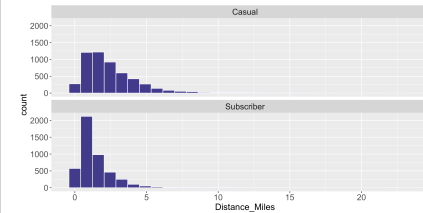
Summarizing Data

	RoutelD	PaymentPlan	StartHub	Distance_Miles
25	3596434	Subscriber	NW 18th at Flanders	0.59
26	3607170	Subscriber	NW Raleigh at 21st	0.71
27	3631639	Casual	SW 2nd at Pine	3.15
28	3912181	Casual	SE Water at Taylor	5.89
29	4031739	Casual	SE Clay at Water	1.34
30	3859969	Casual	SW Naito at Morrison	1.56
31	4315016	Casual	NW Everett at 22nd	3.50
32	4252609	Casual	NW Flanders at 14th	1.81
33	3809564	Casual	NA	2.74

- Hard to do by eyeballing a spreadsheet with many rows!

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Summarizing Data Visually



For a quantitative variable, often want to answer:

- What is an **average** value (the center)?
- What is the **trend/shape** of the variable?
- How much **variation** is there from case to case (spread)?

Q: I will sometimes ask you to assess these qualitatively by eye. Let's practice with these histograms.

To be more precise in assessing these we need to learn key **summary statistics**: Numerical values computed based on the observed cases.

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Measures of Center

Mean: Average of all the observations

- n = Number of cases (sample size)
- x_i = value of the i -th observation
- Denote by $\bar{x}$

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Measures of Center

Mean: Average of all the observations

- n = Number of cases (sample size)
- x_i = value of the i -th observation
- Denote by $\bar{x}$

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

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Measures of Center

Mean: Average of all the observations

- n = Number of cases (sample size)
- x_i = value of the i -th observation
- Denote by $\bar{x}$

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

```
1 # Test out on first 6 values
2 head(biketown$Distance_Miles)
[1] 1.91 0.72 3.42 1.81 4.51 5.54
```

```
1 (1.91 + 0.72 + 3.42 + 1.81 + 4.51 + 5.54) / 6
[1] 2.985
```

```
1 # Compute in R for all values
2 mean(biketown$Distance_Miles)
[1] 2.044768
```

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Measures of Center

Median: Middle value

- Half of the data falls below the median
- Denote by m
- If n is even, then it is the average of the middle two values

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Measures of Center

Median: Middle value

- Half of the data falls below the median
- Denote by m
- If n is even, then it is the average of the middle two values

Compute in R across all values:

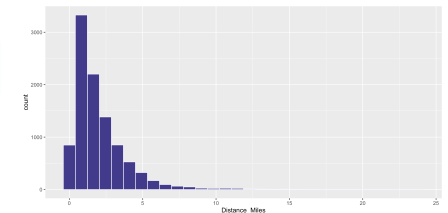
```
1 median(biketown$Distance_Miles)
[1] 1.48
```

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Measures of Center

Q: Why is the mean larger than the median?

```
1 biketown %>%
2   summarize(mean_miles = mean(Distance_Miles),
3             median_miles = median(Distance_Miles))
#> # A tibble: 1 x 2
#>   mean_miles median_miles
#>   <dbl>         <dbl>
#> 1  2.044768         1.48
```



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Measures of Center

Note: The mean is very sensitive to outliers, while the median is not.

```
1 my_data <- c(1, 2, 5, 7, 8, 10)
2 my_data_with_outlier <- c(1, 2, 5, 7, 8, 100)
```

- Q: How will the means compare?

```
1 mean(my_data)           1 mean(my_data_with_outlier)
[1] 5.5                    [1] 20.5
```

- Q: How will the medians compare?

```
1 median(my_data)         1 median(my_data_with_outlier)
[1] 6                      [1] 6
```

We call the median a **robust** statistic.

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Summary statistic anecdote

In April 2026 the New York Times published an article titled “In Defense of Dumb Dogs.” The article describes research that found that a large majority (66%) of dog owners describe their dog as “smarter than average.”

NYT’s morning newsletter stated that this result was “**statistically impossible**.” Apparently many readers wrote in to complain, because the next morning’s newsletter included a (somewhat defensive) mea culpa about this flawed statement.

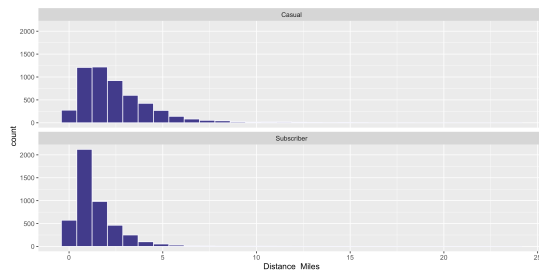
- Q: Why was the NYT newsletter statement flawed?



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Computing Measures of Center by Groups

Question: Who travels further, on average? Casual biketown users or payment plan subscribers?



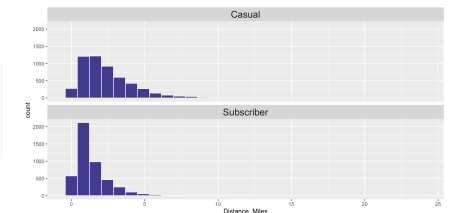
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Computing Measures of Center by Groups

Handy **dplyr** functions that we'll learn: **group_by()** and **summarize()**.

```
1 library(dplyr)
2
3 # Calculate group *MEANS*
4 biketown %>%
5   group_by(PaymentPlan) %>%
6   summarize(mean_dist = mean(Distance_Miles))
# A tibble: 2 x 2
  PaymentPlan mean_dist
  <chr>         <dbl>
1 Casual       2.56
2 Subscriber    1.45
```

```
1 # Calculate group *MEDIAN*
2 biketown %>%
3   group_by(PaymentPlan) %>%
4   summarize(median_dist = median(Distance_Miles))
# A tibble: 2 x 2
  PaymentPlan median_dist
  <chr>         <dbl>
1 Casual       2.03
2 Subscriber    1.02
```



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Measures of Variability

- Want a statistic that summarizes how much observations **deviate** from the mean
- **Idea**:
 - Find how much each observation deviates from the mean.
 - Compute the average of the deviations

$$\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})$$

- Test out on 5 values: 5, 10, 20, 5, 25

Problem?

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Measures of Variability

- Want a statistic that captures how much observations **deviate** from the mean

NEW proposal:

- Find how much each observation deviates from the mean.
- Compute the average of the **squared** deviations.
 - Why?

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Measures of Variability

- Want a statistic that captures how much observations **deviate** from the mean

NEW proposal:

- Find how much each observation deviates from the mean.
- Compute the average of the **squared** deviations.
 - Why?

```
1 # Test out on first 6 values
2 head(biketown$Distance_Miles)
[1] 1.91 0.72 3.42 1.81 4.51 5.54
```

```
1 # Calculate mean by hand
2 (1.91 + 0.72 + 3.42 + 1.81 + 4.51 + 5.54) / 6
[1] 2.985
```

```
1 # Calculate average squared deviations by hand
2 ((1.91 - 2.985)^2 + (0.72 - 2.985)^2 +
3  (3.42 - 2.985)^2 + (1.81 - 2.985)^2 +
4  (4.51 - 2.985)^2 + (5.54 - 2.985)^2) / 6
[1] 2.784892
```

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Measures of Variability

- Want a statistic that captures how much observations **deviate** from the mean

NEW proposal, formula:

- Find how much each observation deviates from the mean.
- Compute the (nearly) average of the **squared** deviations.
- Called **sample variance** s^2 .

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Measures of Variability

- Want a statistic that captures how much observations **deviate** from the mean

NEW proposal, formula:

- Find how much each observation deviates from the mean.
- Compute the (nearly) average of the **squared** deviations.
- Called **sample variance** s^2 .

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

- We'll learn why we divide by $n - 1$ instead of n later in the course.

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Measures of Variability

- Want a statistic that captures how much observations **deviate** from the mean

NEW proposal, formula:

Compute with R:

```
1 var(biketown$Distance_Miles)
[1] 3.805638
```

- Find how much each observation deviates from the mean.
- Compute the (nearly) average of the **squared** deviations.
- Called **sample variance** s^2 .

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

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Measures of Variability

- Want a statistic that captures how much observations **deviate** from the mean

- We found how much each observation deviates from the mean.

Compute in R:

```
1 # Variance
2 var(biketown$Distance_Miles)
[1] 3.805638
```

- Computed the (nearly) average of the **squared** deviations (s^2).

- But because observations are squared, units differ from original data.

```
1 # Standard deviation
2 sqrt(var(biketown$Distance_Miles))
[1] 1.950804
```

- The square root of the sample variance is called the **sample standard deviation** s .

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

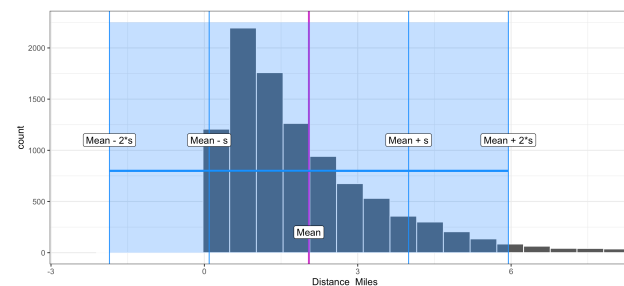
```
1 # Standard deviation (built-in)
2 sd(biketown$Distance_Miles)
[1] 1.950804
```

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Visualizing Standard Deviation

- The standard deviation measures the **typical size of deviations from the mean**.
- For most data sets, the large majority of observations are within 2 standard deviations of the mean.

```
1 sd(biketown$Distance_Miles)
[1] 1.950804
```



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Measures of Variability

- In addition to the sample standard deviation and the sample variance, there is the sample **interquartile range** (IQR):

$$\text{IQR} = Q_3 - Q_1$$

- Imagine we sort our data and divide it into four evenly sized chunks (**quartiles**)
- 25% of all observations are less than the *first quartile* Q_1
- 25% of all observations are greater than the *third quartile* Q_3

Compute with R:

```
1 IQR(biketown$Distance_Miles)
[1] 1.89
```

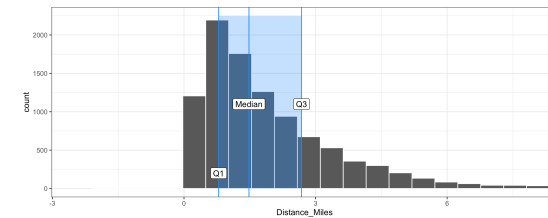
25

Visualizing the IQR

- In addition to the sample standard deviation and the sample variance, there is the sample **interquartile range** (IQR):

$$\text{IQR} = Q_3 - Q_1$$

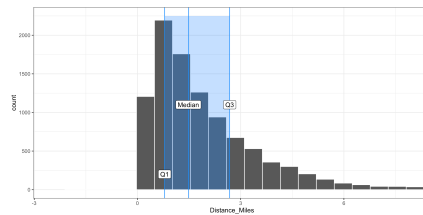
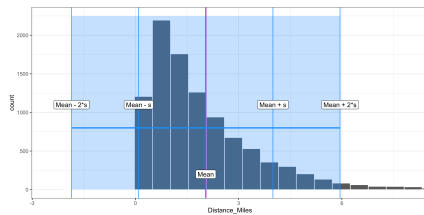
```
1 quantile(biketown$Distance_Miles, c(0.25, 0.5, 0.75))
25% 50% 75%
0.79 1.48 2.68
```



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Comparing Measures of Variability

- Q: Which is more robust to outliers, the IQR or s ?**



- Q: Which is more commonly used, the IQR or s ?**

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Summarizing Categorical Variables

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New data set: Cambridge dogs

```
1 dogs <- read.csv("https://data.cambridgema.gov/api/views/sckh-3xyx/rows.csv")
2 glimpse(dogs)

Rows: 27,774
Columns: 10
$ Dog_Name      <chr> "Poppy Grace", "Clementine", "Momo", "Maple", "MICKEY..."
$ Dog_Breed     <chr> "Pug", "Labrador Retriever", "Pomeranian", "Golden Re..."
$ Location_masked <chr> "POINT (-71.093036 42.371026)", "POINT (-71.115351 42..."
$ Latitude_masked <dbl> 42.3710, 42.3807, 42.3725, 42.3807, 42.3682, 42.3852, ...
$ Longitude_masked <dbl> -71.0930, -71.1154, -71.1084, -71.1154, -71.0791, -71...
$ Neighborhood  <chr> "Wellington-Harrington", "Baldwin", "Mid-Cambridge", "...
$ Birth.Year     <chr> "01/01/2017 12:00:00 AM", "01/01/2015 12:00:00 AM", "...
$ Gender         <chr> "Female", "Female", "Male", "Female", "Male", "Female..."
$ Expiration.Date <chr> "01/01/2021 12:00:00 AM", "01/01/2024 12:00:00 AM", "...
$ Is.Active      <chr> "false", "false", "false", "true", "false", "false", ...
```

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Cambridge dogs data set

May want to focus on the dogs with the 5 most common names

```
1 dogs <- read.csv("https://data.cambridgema.gov/api/views/sckh-3xyx/rows.csv")
2
3 # Useful wrangling that we will come back to
4 dogs_top5 <- dogs %>%
5   mutate(Breed = case_when(Dog_Breed == "Mixed Breed" ~ "Mixed",
6                             TRUE ~ "Single")) %>%
7   filter(Dog_Name %in% c("Luna", "Charlie", "Lucy", "Cooper", "Rosie"))
8
9
10 head(dogs_top5)
  Dog_Name      Dog_Breed
1   Lucy      Keeshond / Dachshund
2   Luna  Labrador Retriever / Rat Terrier
3   Luna  Labrador Retriever / German Shepherd Dog
4  Charlie  Siberian Husky / Labrador Retriever
5   Luna      French Bulldog
6   Luna  Labrador Retriever
  Location_masked Latitude_masked Longitude_masked Neighborhood
1 POINT (-71.096823 42.365754)      42.3658      -71.0968      The Port
2 POINT (-71.13607 42.395315)      42.3953      -71.1361 North Cambridge
3 POINT (-71.108359 42.372506)      42.3725      -71.1084 Mid-Cambridge
4 POINT (-71.152822 42.38067)      42.3807      -71.1528 Strawberry Hill
5 POINT (-71.136168 42.376682)      42.3767      -71.1362 West Cambridge
6 POINT (-71.152822 42.38067)      42.3807      -71.1528 Strawberry Hill
```

30

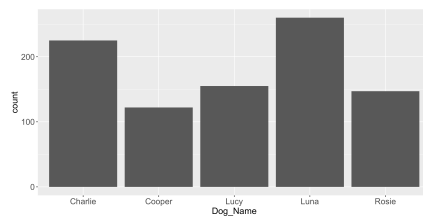
Frequency Table

Distributions of categorical variables can be presented in tables and summarized in bar charts.

```
1 count(dogs_top5, Dog_Name) # Handy function

Dog_Name  n
1 Charlie 225
2 Cooper  122
3 Lucy   155
4 Luna   260
5 Rosie  147
```

```
1 ggplot(data = dogs_top5,
2       mapping = aes(x = Dog_Name)) +
3   geom_bar()
```



Q: Why can't we use mean or standard deviation here?

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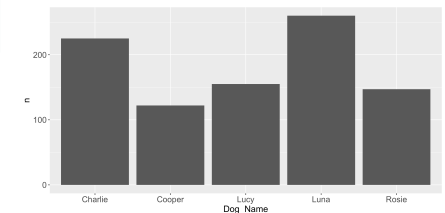
Another ggplot2 geom: geom_col()

If you have already aggregated the data, you will use `geom_col()` instead of `geom_bar()` to make a barplot.

```
1 dog_counts <- count(dogs_top5, Dog_Name)
2 dog_counts
```

```
Dog_Name  n
1 Charlie 225
2 Cooper  122
3 Lucy   155
4 Luna   260
5 Rosie  147
```

```
1 ggplot(data = dog_counts,
2       mapping = aes(x = Dog_Name, y = n)) +
3   geom_col()
```



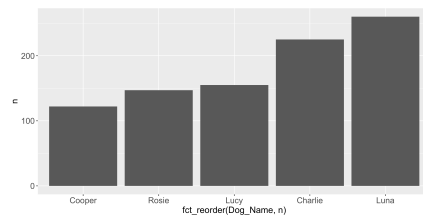
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Another ggplot2 geom: geom_col()

And you can use `fct_reorder` to order bars by value

```
1 dog_counts <- count(dogs_top5, Dog_Name)
2 dog_counts
3
4 Dog_Name  n
5 Charlie 225
6 Cooper  122
7 Lucy   155
8 Luna   260
9 Rosie  147
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```

```
1 library(forcats)
2 ggplot(data = dog_counts,
3       mapping = aes(x = fct_reorder(Dog_Name, n),
4                               y = n)) +
5   geom_col()
```



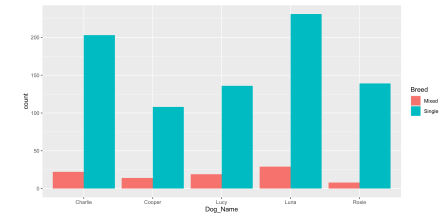
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Contingency Table

To compare 2 categorical variables, we can use a contingency table*

```
1 # Using `count()` from dplyr
2 count(dogs_top5, Dog_Name, Breed)
3
4 Dog_Name Breed  n
5 Charlie Mixed  22
6 Charlie Single 203
7 Cooper Mixed  14
8 Cooper Single 108
9 Lucy Mixed   19
10 Lucy Single 136
11 Luna Mixed   29
12 Luna Single 231
13 Rosie Mixed   8
14 Rosie Single 139
```

```
1 ggplot(data = dogs_top5,
2       mapping = aes(x = Dog_Name, fill = Breed)) +
3   geom_bar(position = "dodge")
```



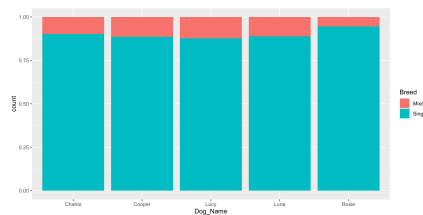
- * (AKA a two-way table, often presented horizontally rather than all vertically)

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Conditional Proportions

- Beyond raw counts, we often summarize categorical data with **conditional** proportions.
 - Especially when looking for relationships!

```
1 ggplot(data = dogs_top5,
2       mapping = aes(x = Dog_Name, fill = Breed)) +
3   geom_bar(position = "fill")
```



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Conditional Proportions

```
1 count(dogs_top5, Dog_Name, Breed)
2
3 Dog_Name Breed  n
4 Charlie Mixed  22
5 Charlie Single 203
6 Cooper Mixed  14
7 Cooper Single 108
8 Lucy Mixed   19
9 Lucy Single 136
10 Luna Mixed   29
11 Luna Single 231
12 Rosie Mixed   8
13 Rosie Single 139
```

```
1 count(dogs_top5, Dog_Name, Breed) %>%
2   group_by(Dog_Name) %>%      ## Conditions on Dog Name
3   mutate(prop = n / sum(n))  ## Adds/changes columns
4
5 # A tibble: 10 x 4
6 # Groups:   Dog_Name [5]
7   Dog_Name Breed  n prop
8   <chr>    <chr> <int> <dbl>
9 1 Charlie Mixed  22 0.0978
10 2 Charlie Single 203 0.902
11 3 Cooper Mixed  14 0.115
12 4 Cooper Single 108 0.885
13 5 Lucy Mixed   19 0.123
14 6 Lucy Single 136 0.877
15 7 Luna Mixed   29 0.112
16 8 Luna Single 231 0.888
17 9 Rosie Mixed   8 0.0544
18 10 Rosie Single 139 0.946
```

We'll go over these data wrangling functions on Wednesday!

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Conditional Proportions

```
1 count(dogs_top5, Dog_Name, Breed) %>%
2 group_by(Dog_Name) %>% ## Conditions on DOG NAME
3 mutate(prop = n / sum(n)) ## Adds/changes columns
```

```
# A tibble: 10 × 4
# Groups:   Dog_Name [5]
  Dog_Name Breed      n prop
  <chr>    <chr> <int> <dbl>
1 Charlie Mixed     22 0.0978
2 Charlie Single    203 0.902
3 Cooper  Mixed     14 0.115
4 Cooper  Single    108 0.885
5 Lucy    Mixed     19 0.123
6 Lucy    Single    136 0.877
7 Luna    Mixed     29 0.112
8 Luna    Single    231 0.888
9 Rosie   Mixed      8 0.0544
10 Rosie   Single    139 0.946
```

```
1 count(dogs_top5, Dog_Name, Breed) %>%
2 group_by(Breed) %>% ## Conditions on BREED
3 mutate(prop = n / sum(n)) ## Adds/changes columns
```

```
# A tibble: 10 × 4
# Groups:   Breed [2]
  Dog_Name Breed      n prop
  <chr>    <chr> <int> <dbl>
1 Charlie Mixed     22 0.239
2 Charlie Single    203 0.248
3 Cooper  Mixed     14 0.152
4 Cooper  Single    108 0.132
5 Lucy    Mixed     19 0.207
6 Lucy    Single    136 0.166
7 Luna    Mixed     29 0.315
8 Luna    Single    231 0.283
9 Rosie   Mixed      8 0.0870
10 Rosie   Single    139 0.170
```

Q: What's different between these two tables?

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Recap: quantitative data

- We'll frequently **summarize** quantitative data by assessing a distribution's:
 1. Center
 2. Spread (aka variability)
 3. Shape
- If we want to assess these visually, we should use a boxplot or histogram
- We quantify center and spread by calculating (or sometimes by eyeballing) **summary statistics**
 1. The **mean** or **median** (`mean()`, `median()`) for center
 2. The **standard deviation**, **variance**, or **interquartile range** (`sd()`, `var()`, `IQR()`) for spread
- In this class we'll usually assess shape visually (**symmetric**, **right-skewed**, or **left-skewed**?)
 - We'll see a quantitative example on the homework

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Recap: categorical data

- We'll frequently **summarize** categorical data by making **tables** of category **counts** or **proportions**
 - Table of *counts* for 1 categorical variable = **frequency table** or **one-way frequency table**
 - Table of *counts* for 2 categorical variables = **contingency table** or **two-way frequency table**
 - Table of *proportions* with 2 categorical variables = **conditional proportions table**
- We can most easily visualize these tables via barplots

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Reminders

- The teaching team would love to see you in office hours! Check Moodle for the schedule.

Next time:

- We'll define **data wrangling** and
- Learn to use more functions and methods to summarize and wrangle data

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